

AIG130 – Project 2

Introduction

This project builds upon a solution you have developed in any of AIG ML related labs/projects OR any ML project you have done, incorporating Docker to containerize the machine learning pipeline. By using Docker, you will create a portable and consistent environment for deploying, testing, and scaling your solution. The focus will be on leveraging containerization to enhance reproducibility, collaboration, and scalability.

Objective

- Adapt an existing solution into a **Dockerized environment**.
- Understand and implement best practices for containerizing machine learning applications.
- Evaluate the impact of containerization on deployment, reproducibility, and scalability.
- Use Docker to manage dependencies, configurations, and runtime environments for your solution.

Group Formation

Students can work individually or form groups of up to 4 members. Collaboration and effective communication are encouraged for those working in groups.

Assignment

1) Pipeline Review and Enhancement

- a) Review the machine learning pipeline you intend to containerize.
- b) Identify any enhancements required for seamless containerization (e.g., managing external dependencies, ensuring code modularity).

2) Containerization with Docker

- a) Write a **Dockerfile** to containerize your machine learning pipeline. Include:
 - i) The environment setup (e.g., Python version, required libraries).
 - ii) Installation of any additional library that your pipeline requires.

3) Testing and Validation

- a) Test the Dockerized solution to ensure:
 - i) It runs identically across different environments.
 - ii) Dependencies and configurations are correctly encapsulated.
 - iii) Integration points (e.g., pre-trained models or datasets from HuggingFace) function as expected.

Deliverables

Report: Submit a PDF report covering all tasks. Include:

- An introduction and conclusion.

- Sections for each task with headings.
- Visual elements (charts, tables) where applicable.

Demo: Prepare a video demo/presentation of your implemented solution.

- Duration: Up to 15 minutes.
- Demonstrate the part(s) you implemented. For the remaining parts, just provide an explanation.

Individual Reflection

Each student must submit a separate reflection document answering the following:

1. If you worked individually, why did you choose to go solo? What challenges did you encounter?
2. If you did it as a group, how were tasks distributed, and what was your specific task? What challenges did you encounter?
3. Explain in your own words the advantages and disadvantages of using cloud services for implementing and deploying machine learning applications. You can use the application you selected for this project as a reference in your answer.

Submission

Submissions are done through the provided links in Blackboard. There are three links:

1. Group report (one submission per group): 8%
2. Demo (one submission per group): 8%
3. Individual reflection: 4%

Assessment Criteria

- Quality and depth of research, analysis, and implementation.
- Clarity and organization of the report.
- Quality and clarity of the demo.
- Thoughtfulness of the individual reflection.